

STEPHENVILLE CLARK, TX, USA

WMO: 722600

Lat: 32.215N

Lon: 98.178W

Elev: 403

StdP: 96.58

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 03969

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-5.1	-2.8	-13.1	1.3	3.8	-11.2	1.5	4.5	11.1	8.3	10.1	7.7	3.1	350	0.433

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	12.1	37.6	21.4	36.4	21.6	35.2	21.7	24.2	31.0	23.8	30.6	23.4	30.2	3.4	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.5	18.0	26.8	22.2	17.8	26.5	21.7	17.2	26.0	75.3	31.0	73.2	30.6	71.6	30.0	35.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.3	8.3	7.5	DB	-8.8	39.2	2.2	1.3	-10.4	40.2	-11.7	41.0	-12.9	41.7	-14.5	42.7	(4)
(5)			WB	-9.7	25.9	2.0	2.7	-11.2	27.8	-12.4	29.3	-13.5	30.8	-15.0	32.8	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	18.6	8.0	9.9	14.6	18.3	22.4	26.7	28.3	24.8	24.8	19.2	13.5	8.8	(6)	
	DBStd	8.44	5.48	5.80	5.17	4.20	3.62	2.44	2.04	2.48	3.15	4.60	5.15	5.26	(7)	
	HDD10.0	308	103	68	18	3	0	0	0	0	0	3	25	87	(8)	
	HDD18.3	1258	322	241	134	49	11	0	0	0	1	45	157	297	(9)	
	CDD10.0	3451	41	64	162	252	385	501	566	568	443	288	129	50	(10)	
	CDD18.3	1359	1	4	20	49	137	251	308	310	193	72	12	2	(11)	
	CDH23.3	14948	22	56	191	454	1201	2777	3734	3771	1932	681	104	26	(12)	
	CDH26.7	7271	2	12	41	131	442	1374	2046	2087	888	226	19	3	(13)	
(14) Wind	WSAvg	3.1	3.3	3.6	3.9	4.0	3.5	3.1	2.5	2.3	2.3	2.6	3.1	3.2	(14)	
(15) Precipitation	PrecAvg	876	48	62	73	78	99	104	53	77	75	82	62	55	(15)	
	PrecMax	1306	93	233	193	192	221	304	124	234	174	215	205	247	(16)	
	PrecMin	524	13	1	2	5	48	30	0	4	9	18	7	3	(17)	
	PrecStd	218	23	54	45	50	44	70	38	56	43	50	52	54	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	26.2	28.0	30.1	32.7	34.8	37.7	39.2	39.1	37.2	33.1	29.1	26.3	(19)	
		MCWB	13.6	15.3	16.8	18.5	21.2	21.4	21.0	21.3	20.4	20.3	18.2	15.7	(20)	
	2%	DB	22.6	24.9	27.7	30.0	32.5	36.2	37.5	37.6	34.9	31.2	26.3	22.8	(21)	
		MCWB	12.8	15.1	16.7	18.1	21.1	21.4	21.4	21.5	21.1	20.5	16.7	13.8	(22)	
	5%	DB	20.0	22.3	25.6	27.7	31.1	34.8	36.2	36.7	33.0	29.1	23.9	20.6	(23)	
		MCWB	12.6	13.9	16.0	18.0	20.7	21.6	21.7	21.8	21.1	19.2	16.0	13.3	(24)	
	10%	DB	17.4	19.6	22.9	26.1	29.2	32.8	34.9	35.2	32.0	27.4	22.1	17.7	(25)	
		MCWB	11.3	13.2	15.1	17.4	20.5	21.9	21.9	21.9	21.1	17.9	15.3	12.2	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	17.9	18.9	20.1	21.9	24.1	25.2	25.0	24.6	24.1	23.0	20.6	18.8	(27)	
		MCDB	20.1	23.4	25.5	26.6	30.4	31.8	32.2	31.4	30.3	28.5	25.3	22.2	(28)	
	2%	WB	15.8	17.5	19.0	20.8	23.1	24.1	24.1	24.0	23.5	22.1	19.0	17.1	(29)	
		MCDB	19.3	21.3	23.9	26.0	28.9	30.6	30.9	31.2	29.6	27.5	22.8	20.4	(30)	
	5%	WB	14.0	16.0	18.1	20.1	22.4	23.5	23.6	23.6	22.9	21.4	18.0	15.1	(31)	
		MCDB	17.8	19.9	22.5	25.1	28.0	30.0	30.6	30.9	28.9	26.2	21.5	18.0	(32)	
	10%	WB	11.7	14.1	17.2	19.2	21.8	22.9	23.1	23.0	22.4	20.4	17.0	12.9	(33)	
		MCDB	16.6	18.4	21.1	23.9	27.1	29.4	30.2	30.6	28.2	24.9	20.6	17.2	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	12.4	12.2	12.6	12.4	11.2	11.4	12.1	12.1	11.8	12.7	12.4	12.2	(35)	
		MCDBR	16.2	16.0	15.0	14.8	13.4	13.7	14.1	14.0	13.6	14.2	14.6	16.4	(36)	
	5% WB	MCWBR	8.9	8.6	6.7	6.6	4.9	3.3	3.2	3.2	3.8	5.7	7.4	9.2	(37)	
		MCWBR	12.7	13.5	11.3	11.3	10.6	10.8	11.4	11.7	10.8	11.3	11.3	12.4	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.302	0.309	0.339	0.375	0.409	0.414	0.423	0.414	0.388	0.351	0.324	0.304	(40)	
		taud	2.524	2.512	2.443	2.364	2.331	2.352	2.372	2.376	2.394	2.485	2.492	2.523	(41)	
	Ebn at Noon		927	952	942	914	881	872	862	867	878	893	890	904	(42)	
		Edh at Noon	84	94	109	123	128	125	122	120	113	96	87	81	(43)	
(44) All-Sky Solar Radiation	RadAvg		3.04	3.62	4.72	5.70	6.05	6.79	6.85	6.24	5.25	4.16	3.21	2.69	(44)	
	RadStd		0.32	0.55	0.37	0.48	0.59	0.58	0.46	0.42	0.43	0.63	0.41	0.35	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (49 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page